

Week 14: Relations (4.2)

Informally: a "relation" is a way to compare "things"

Example: $S = \mathbb{R}$, $\geq$ is a relation

$\forall a, b \in S$, " $a \geq b$ " is either true or false

Defn: let S be a set. A relation on S is a subset $R \subseteq S \times S$.

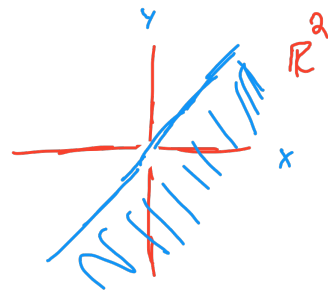
Usually use $\sim$

If $(a, b) \in R$, we say that " a is related to b " and write $a \sim b$ (or $a \sim_R b$).

Example: $S = \mathbb{R}$

$R \subseteq \mathbb{R} \times \mathbb{R}$ given by

$$R \stackrel{\text{def}}{=} \{(a, b) \in \mathbb{R} \times \mathbb{R} \text{ s.t. } a \geq b\}$$



Example: $S = \{0, 1, 2\}$

$$R \subseteq S \times S$$

$$R \stackrel{\text{def}}{=} \{(0, 1), (0, 2), (1, 2)\}$$

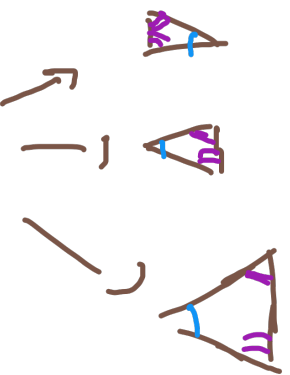
$0 \sim 1$ true b/c $(0, 1) \in R$

$1 \sim 2$ true

$2 \sim 0$ false b/c $(2, 0) \notin R$

"Equivalence" Relations

Axiomatize the notion of "the same" or "more or less the same"
there are many properties of Δ 's which don't depend on
size or orientation.



basically the same
not literally the same

Example: ①

$$\frac{2}{3} = \frac{4}{6}$$

Defn: Let S be a set.. Let R be a relation on S

We say that R is an equivalence relation if

$$(R) \forall a \in S, a \sim a$$

• Reflexive

$$(S) \forall a, b \in S, a \sim b \Rightarrow b \sim a$$

• Symmetric

$$(T) \forall a, b, c \in S, a \sim b \wedge b \sim c \Rightarrow a \sim c$$

• transitive

Let $a \in S$. We define the equivalence class of a to be

$$[a] \stackrel{\text{def}}{=} \{b \in S \text{ s.t. } b \sim a\}$$

NOTE: if $b \in [a]$ then $[b] = [a]$

$\uparrow$ $\uparrow$
Set Sets

Warning:

Not interval notation.

$$a \in [a] \quad b \in a \sim a$$

new notation $\rightarrow$

$$[a] \text{ vs } \{a\}$$

$\uparrow$

$\uparrow$ the set whose only elt is a

more
elts

Examples: $S = \mathbb{Z}$, $a \sim b$ if $a \mid a-b$

$$2 \sim 4 \text{ b/c } 2-4 = -2 \text{ and } 2 \mid -2$$

$$1 \sim 3 \quad 1-3 = -2$$

$$1 \not\sim 2 \quad 1-2 = -1 \quad 2 \nmid -1$$

(i.e., $a \sim b$ if a and b have the same remainder when you divide by 2)

(If a and b have the same parity)

$$\begin{aligned} (2,4) &\in R \\ (2,3) &\notin R \end{aligned}$$

Claim: this is an equiv. relation.

Pf: (R) Let $a \in \mathbb{Z}$. (WTP: $a \sim a$, i.e., $2 \mid a-a$)

Since $a-a=0$, $2 \mid a-a$, so $a \sim a$.

(S) Let $a, b \in \mathbb{Z}$. Suppose $a \sim b$. Then $2 \mid a-b$.

(WTP: $b \sim a$, i.e., $2 \mid b-a$). Since $b-a = -(a-b)$,

$2 \mid b-a$, so $b \sim a$.

(T) Let $a, b, c \in \mathbb{Z}$. Suppose $a \sim b$ and $b \sim c$. Then $2 \mid a-b$ and $2 \mid b-c$.

(WTS: $a \sim c$, i.e., $2 \mid a-c$) then $2 \mid (a-b) + (b-c)$, so $2 \mid a-c$. Thus $a \sim c$. $\square$

Q: What are the equiv. classes?

$$[0] = \{a \in \mathbb{Z} \text{ s.t. } a \sim 0\}$$

$$= \{a \in \mathbb{Z} \text{ s.t. } 2 \mid a\}$$

$$= 2\mathbb{Z} = \mathbb{E}$$

$$a \sim 0 \iff 2 \mid a-0$$

$$\iff 2 \mid a$$

$$[1] = \{a \in \mathbb{Z} \text{ s.t. } a \sim 1\}$$

$$= \{a \in \mathbb{Z} \text{ s.t. } a \text{ is odd}\}$$

$$= 2\mathbb{Z}+1 \text{ or } \mathbb{O}$$

$$a \sim 1 \iff 2 \mid a-1$$

$$\iff a-1 \text{ is even}$$

$$\iff a \text{ is odd}$$

NOTE: $[0] \cup [1] = \mathbb{Z}$ AND $[0] \cap [1] = \emptyset$ "partition"

$$[2] = \{a \in \mathbb{Z} \text{ s.t. } a \sim 2\}$$

$$= \mathbb{E}$$

$$a \sim 2 \iff 2 \mid a-2$$

$$\iff 2 \mid a$$

$$[0] = [2]$$

$$[0] \neq [1]$$

$$0 \in [0] \text{ but } 0 \notin [1]$$

$S = \mathbb{R}$ $x \sim y$ if $x < y$

NOT AN E.R. h/c

Not (R) or (S). (T.S (T))

Pf: Let $a=0$. Then $0 < 0$ is false, so $a \not\sim 0$.

Thus $<$ is not reflexive.

Let $a=0$ and $b=1$. Then $0 < 1$, so $0 \sim 1$, but $1 \not\sim 0$, so not.

(T) $a \times b \wedge b \times c \Rightarrow a \times c$.

$S = \mathbb{R}$, $x, y \in \mathbb{R}$, $x \sim y$ if $x - y \in \mathbb{Q}$

$$\pi \sim \pi + 1 \quad \pi - (\pi + 1) = -1 \in \mathbb{Q}$$

$$0 \neq \sqrt{2} \quad \text{h.c.} \quad \sqrt{2} - 0 = \sqrt{2} \notin \mathbb{Q}$$

Claim: there is an $E \not\subseteq \mathbb{R}$.

Pf. (P) let $a \in \mathbb{R}$. Then $a - a = 0 \in \mathbb{Q}$. Thus $a \sim a$.

(S) let $a, b \in \mathbb{R}$. Suppose $a \sim b$, i.e., $a - b \in \mathbb{Q}$.

Then $b - a \in \mathbb{Q}$, so $b \sim a$.

(T) let $a, b, c \in \mathbb{R}$. Suppose $a \sim b$ and $b \sim c$, i.e., $a - b \in \mathbb{Q}$ and $b - c \in \mathbb{Q}$.

Adding gives $a - c = (a - b) + (b - c) \in \mathbb{Q}$, thus $a \sim c$. $\square$

A, B sets. Define $A \sim B$ if $\exists$ a bijection $f: A \rightarrow B$.

Thm f is bi $\iff f$ has an inverse

$$\{1\} \sim \{2\} \text{ via } f(1) = 2$$

$$\{1\} \not\sim \{1, 2\}$$

Claim: This is an E.R.

Pf: (R) Let A be a set. (wrt: $A \sim A$, i.e., $\exists$ a bijection $A \xrightarrow{f} A$)

Then $f = \text{id}_A$ is a bijection from A to A , hence the inverse of id_A is id_A , i.e., $\text{id}_A \circ \text{id}_A = \text{id}_A$.

(S) Let A, B be sets. Suppose $\exists$ a bijection $f: A \rightarrow B$.

By the Thm, $\exists g: B \rightarrow A$ s.t. $g = f^{-1}$. Since g is invertible, g is a bijection.

(T) Let A, B, C be sets. Suppose $\exists f: A \rightarrow B$ and $g: B \rightarrow C$ s.t.

f and g are bijective. Then $g \circ f: A \rightarrow B$ is bijective (b/c we proved in L1d).

Thus $A \sim C$. $\square$

A, B sets, $A \sim B$ if

$$\exists f: A \rightarrow B \text{ s.t.}$$

$$f \text{ is surj.}$$

$$\sim \text{ is } (R) + (T) \text{ (Semi pf)}$$

But: not (S).

$$A = \{1, 3\}, B = \{2, 3\}$$

there are 2 fns from $A \rightarrow B$

$$f(1) = 2$$

$$g(1) = 3$$

Neither is a surjection.

$\exists \text{ surj}$	$B \rightarrow A$
$\nexists \text{ surj}$	$A \rightarrow B$

$S = \mathbb{R}$ $x \sim y$ if $x=1$ or $y=1$

$1 \sim 1$ $2 \not\sim 2$
 $1 \sim 2$ $2 \not\sim 3$
 $2 \sim 1$.

True

Not R) hle $2 \not\sim 3$

(S) is true.

Pf. Let $a, b \in \mathbb{R}$. Sp $a \sim b$, i.e., $a=1$ or $b=1$.

(with: $b \wedge a, i.e., b=1$ or $a=1$) Since "or" is commutative
 $b=1$ or $a=1$. Thus $b \sim a$.

(T) $a=2, b=1, c=3$

then $a \sim b \wedge b \sim c$, but $a \not\sim c$

ER $\Leftrightarrow R \wedge S \wedge T$

$\neg ER \Leftrightarrow \neg R \vee \neg S \vee \neg T$